

Lecture 5: Elastic vs Viscous vs Viscoelastic Behavior

Key Topics

How Materials Deform

Contrasting three fundamental behaviors: 1. Elastic Solids - Store energy, recover shape 2. Viscous Fluids - Dissipate energy, flow continuously 3. Viscoelastic Materials - Combine both behaviors

1. Elastic (Hookean) Behavior

Defining Characteristics

Hooke's Law:

$$\sigma = E \cdot \varepsilon$$

Where: - σ = stress (force per unit area) - E = Young's modulus (material stiffness) - ε = strain (fractional deformation)

Key Properties: - Stress proportional to strain (not rate) - Deform under load but store energy - Recover original shape when load removed - Energy stored elastically (like a spring)

Examples

- Spring
- Rubber band (small extension)
- Steel beam (small deformations)
- Hard plastic below glass transition temperature

Stress-Strain Curves

Linear Elastic Regime: - Straight line - Slope = Young's modulus E - Higher $E \rightarrow$ more stress needed for same strain (stiffer)

Shear Modulus: - For shear deformations: $\tau = G \cdot \gamma$ - G = shear modulus - Similar concept, different geometry

Typical Moduli Values

- Steel: $E \sim 200$ GPa (very stiff)
- Rubber: $E \sim 1$ MPa (soft)
- Silicone gel: $E \sim 50$ kPa (very soft)
- Brain tissue: $E \sim 1$ kPa (extremely soft)

2. Viscous (Newtonian) Behavior

Defining Characteristics

Newton's Law of Viscosity:

$$\tau = \eta \cdot \dot{\gamma}$$

Where: - τ = shear stress - η = viscosity (Pa·s) - $\dot{\gamma}$ = shear rate (strain rate)

Key Properties: - Stress proportional to strain rate (not strain itself) - Flows continuously under any applied stress - Dissipates energy as heat (irreversible) - No shape recovery when stress removed

Examples

- Water ($\eta \approx 1 \text{ mPa}\cdot\text{s}$)
- Honey ($\eta \approx 10 \text{ Pa}\cdot\text{s}$, $\sim 10,000\times$ water)
- Glycerol ($\eta \approx 1 \text{ Pa}\cdot\text{s}$)
- Motor oil

Dimensional Analogy

Property	Elastic	Viscous
Relation	$\sigma = E \cdot \epsilon$	$\tau = \eta \cdot \dot{\gamma}$
Units	$\text{Pa} = \text{N}/\text{m}^2$	$\text{Pa}\cdot\text{s}$
Response to	Strain	Strain rate
Energy	Stored	Dissipated

Behavior Under Load

- If you stop applying stress to viscous fluid:
 - Deformation continues until stress is zero
 - No recovery
 - Fluid remains in deformed state

3. Viscoelastic Behavior

Combined Characteristics

Definition: Materials exhibiting both elastic and viscous behaviors - Deform like elastic solid on short timescales
- Flow like viscous fluid over long times - Behavior depends on observation timescale

Common Viscoelastic Materials

- Polymers (quintessential examples)
- Biological tissues (cartilage, muscle, skin)
- Gels
- Silly putty (classic demo material)
- Many food products (dough, cheese, etc.)

Spring-Dashpot Models

Maxwell Model

Configuration: Spring and dashpot in series

Represents: Fluid with memory

Behavior: - Can relax stress over time - If sudden strain applied and held: - Initially: High stress (spring compressed) - Over time: Stress gradually decays (dashpot flows) - Eventually: Stress $\rightarrow 0$ (full relaxation)

Equation:

$$\dot{\epsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\eta}$$

Relaxation Time:

$$\tau_M = \frac{\eta}{E}$$

Kelvin-Voigt Model

Configuration: Spring and dashpot in parallel

Represents: Solid that creeps under load

Behavior: - Under sudden load: - No instantaneous elastic jump (dashpot resists) - Gradual spring deformation (creep) - Approaches equilibrium asymptotically - When load removed: - Slow recovery back to original shape

Equation:

$$\sigma = E\varepsilon + \eta\dot{\varepsilon}$$

Retardation Time:

$$\tau_K = \frac{\eta}{E}$$

Stress Relaxation and Creep

Stress Relaxation

Experiment: 1. Apply sudden fixed strain 2. Hold constant 3. Measure stress vs time

Results: - Elastic material: Constant stress (no decay) - Viscoelastic (Maxwell): Stress decays exponentially

$$\sigma(t) = \sigma_0 e^{-t/\tau_M}$$

- Time constant τ_M = relaxation time

Creep

Experiment: 1. Apply constant stress 2. Measure strain vs time

Results: - Elastic: Immediate strain, then constant - Viscous: Strain increases linearly forever - Viscoelastic (Kelvin-Voigt): - Immediate dashpot resistance - Gradual approach to equilibrium -

$$\varepsilon(t) = \frac{\sigma_0}{E}(1 - e^{-t/\tau_K})$$

Relaxation Time Concept

Definition

Relaxation Time (τ): Characteristic time for material to transition from elastic to viscous response

Physical Meaning: - Maxwell time: $\tau = \eta/E$ - Time for stress to decay to 1/e of initial value - Separates elastic (short-term) and viscous (long-term) regimes

Time-Scale Dependence

Fast Observation ($t \ll \tau$): - Material behaves elastic (solid-like) - Dashpot “frozen” (hasn’t had time to flow) - Spring dominates

Slow Observation ($t \gg \tau$): - Material behaves viscous (liquid-like) - Spring equilibrated - Dashpot flows freely

Application to Silly Putty: - $\tau \sim$ few seconds - Quick poke ($t \ll \tau$): Bounces like ball (elastic) - Slow pull ($t \gg \tau$): Flows like liquid (viscous)

Examples and Case Studies

Silly Putty Demonstration

Classic Demo:

Fast Impact (High Strain Rate): 1. Roll into ball 2. Drop from height 3. Bounces like elastic ball 4. Behavior: Solid-like ($t \ll \tau$)

Slow Deformation (Low Strain Rate): 1. Take same putty 2. Slowly pull apart 3. Flows and strands out like liquid 4. Can form thin filament that necks 5. Behavior: Liquid-like ($t \gg \tau$)

Very Fast (Brittle Failure): - Quick yank $\rightarrow$ snaps (brittle behavior) - Elastic domination with insufficient time for viscous flow

Metal vs Polymer Comparison

Metal Spring: - Apply weight briefly - Springs back immediately - Pure elastic response - Negligible viscous component

Viscoelastic Polymer Strip: - Apply same weight - Might slowly rebound - Or not fully recover if short time - Clear viscoelastic behavior

Pitch Drop Experiment

Famous Long-Term Experiment:

Short Timescale (Impact): - Block of pitch (tar) - Hit with hammer $\rightarrow$ shatters - Feels solid, elastic

Long Timescale (Years): - Over ~ 10 years, drops form and fall out - Flows like extremely viscous liquid - Demonstrates extreme relaxation time (years!)

Lesson: Time scale of observation matters!

Quantitative Example

Dynamic Testing: - Water: Phase angle $\delta \approx 90^\circ$ (all viscous, no elastic storage) - Rubber: Phase angle $\delta \approx 0^\circ$ (all elastic storage) - Silly putty: Phase angle $\delta \approx 45^\circ$ at intermediate frequencies (half-half)

Units Reminder: - Elastic modulus: Pa - Viscosity: Pa·s - Example values: - Rubber $E \sim 1$ MPa - Honey $\eta \sim 10$ Pa·s - Silly putty $\tau \sim$ seconds

Learning Outcomes

Students will be able to:

1. Differentiate elastic vs viscous response
 - State defining constitutive relations (Hooke's law vs Newton's viscosity law)
 - Qualitatively describe behavior under load
2. Give examples of each type
 - Elastic: Steel, hard plastic below T_g , rubber (small strains)
 - Viscous: Water, oil, glycerin
 - Viscoelastic: Silly putty, cartilage, polymer melt
 - Justify classifications
3. Understand viscoelastic models
 - Maxwell model: Initial stress from spring, then decays as dashpot flows
 - Kelvin-Voigt: Sudden stress $\rightarrow$ dashpot resistance + gradual spring deformation
 - Conceptual understanding (no ODE solving required)
4. Interpret storage vs loss qualitatively
 - Viscoelastic material has:

- Storage modulus (elastic part, energy stored)
 - Loss modulus (viscous part, energy dissipated)
 - More elastic → bounces back (stores energy)
 - More viscous → damps motion (loses energy as heat)
5. Apply time-scale reasoning
- Predict material behavior based on deformation rate
 - Quick stretch → stiff (elastic)
 - Slow stretch → soft/flowy (viscous)
 - Articulate relaxation time concept

Discussion Questions

1. Identify Behaviors

Scenarios to classify:

- Bowling ball impacts cement floor and rebounds
 - Mostly elastic (some energy lost)
- Person slowly sinking into memory foam mattress
 - Viscoelastic (slow deformation over time)
- Toothpaste extruded from tube
 - Mostly viscous flow (with yield stress)
- Guitar string vibrating
 - Elastic

Task: Students debate and justify answers

2. Spring vs Dashpot Analogy

Questions:

- If you stretch Maxwell material (spring+dashpot series) and hold, what happens to stress?
 - Initially: High stress from spring
 - Over time: Gradually decays (dashpot relieves it)
 - Stress relaxation
- If suddenly released after long hold, will it return to original length?
 - No – dashpot has flowed (permanent deformation)
 - Only spring part returns
 - Some deformation remains
- How does this differ from Kelvin-Voigt under sudden load?

3. Energy Storage vs Loss

Scenario: Two materials feel similarly stiff when tapped, but one gets warm when bent repeatedly, the other doesn't.

Question: What does this indicate? - Warming material: Dissipating energy → higher loss modulus - Non-warming: Stores-returns energy → higher storage modulus

Examples: - Rubber bands heating up after stretching - Car tires heating up (viscoelastic damping)

4. Design Considerations

Shoe Sole for Runners: - More elastic or more viscous? - Discussion: - Elastic: Returns energy (spring in step) - Viscous: Absorbs energy (shock absorption) - Likely want mix (viscoelastic) - Different answers possible - Shows engineering balance

Seismic Dampers: - Should be highly viscous - Dissipate earthquake energy - Not elastic (would spring back and forth) - Connects material behavior to function

Key Concepts to Remember

Elastic Behavior

- Relation: $\sigma = E \cdot \epsilon$
- Energy: Stored (recoverable)
- Response: Immediate
- Examples: Springs, metals, rubber (small strain)

Viscous Behavior

- Relation: $\tau = \eta \cdot \dot{\gamma}$
- Energy: Dissipated as heat
- Response: Continuous flow
- Examples: Water, honey, oils

Viscoelastic Behavior

- Models: Maxwell (fluid-like) vs Kelvin-Voigt (solid-like)
- Key parameter: Relaxation time $\tau = \eta/E$
- Time dependence: Solid when fast, liquid when slow
- Examples: Polymers, silly putty, tissues

Storage vs Loss

- Storage modulus (G'): Elastic energy stored
- Loss modulus (G''): Viscous energy dissipated
- Phase angle (δ): $\tan \delta = G''/G'$
- Sets up next lecture on rheology!

Preparation for Next Lecture

Next lecture: Rheology – Measuring “Softness”

Consider: - How do we experimentally measure these properties? - What is a rheometer? - How do G' and G'' relate to oscillatory tests? - What does “soft” really mean quantitatively?

References

1. “What is a Viscoelastic material?” - Biolin Scientific
2. Ferry, J.D. “Viscoelastic Properties of Polymers” (1980)
3. Lakes, R.S. “Viscoelastic Materials” (2009)
4. Newtonian fluid - Wikipedia

Previous Lecture: *Lecture 4: Finite-Time Singularities* Next Lecture: *Lecture 6: Rheology* Course Home: *Course Overview*

Metadata: Author:: [Vatsal Sanjay](#) Date published:: Jul 19, 2026 Date modified:: Jul 19, 2026